



9647/01

21 September 2015

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the Multiple Choice Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Multiple Choice Answer Sheet

Write your name, PDG and NRIC / FIN number, **including** the reference letter.

Shade the NRIC / FIN number.

Exam Title: JC2 PRELIM

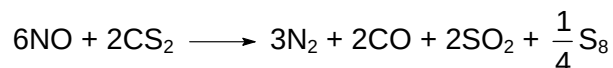
Exam Details: H2 Chemistry / Paper 1

This document consists of **19** printed pages.

Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

- 1 An exciting chemical demonstration is the 'barking dog'. An equation which describes the reaction is shown.

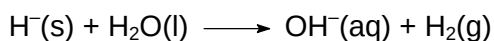


Carbon disulfide, CS_2 , is a liquid with a density of 1.26 g cm^{-3} .

Which mass of sulfur would be formed from 8.00 cm^3 of CS_2 if the reaction proceeded with 100% yield?

- A** 2.67 g **B** 4.25 g **C** 8.49 g **D** 16.99 g

- 2 Group I and Group II ionic hydrides react with water.

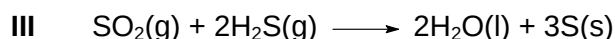
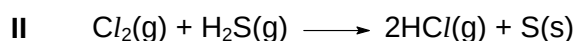
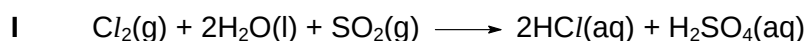


In an experiment, 1 g samples of each of the following ionic hydrides are treated with an excess of water.

Which sample produces the greatest mass of hydrogen?

- A** LiH **B** NaH **C** CaH_2 **D** MgH_2

- 3 The equations for three reactions involving Cl_2 , SO_2 and H_2S are given below.



Which deduction **cannot** be made from the information above?

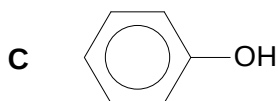
- A** Cl_2 is the weakest reducing agent.
B H_2S is the strongest reducing agent.
C There is an increase in oxidising power in the sequence SO_2 , H_2S , Cl_2 .
D The oxidation number of S changes from -2 in H_2S to 0 in S for reaction **II**.

4 Which atom has the highest ratio of unpaired electrons to paired electrons in its ground state?

- A carbon
- B nitrogen
- C oxygen
- D fluorine

5 Which of the following is a planar molecule?

- A C/F_4^+
- B $\text{CH}_2=\text{C}=\text{CH}_2$



- D $\text{CH}_2=\text{CH}-\text{CHO}$

6 Some car paints contain small flakes of silica, SiO_2 .

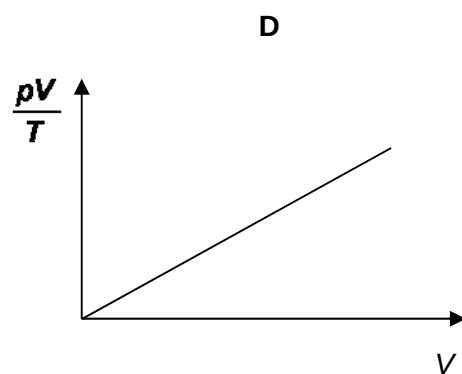
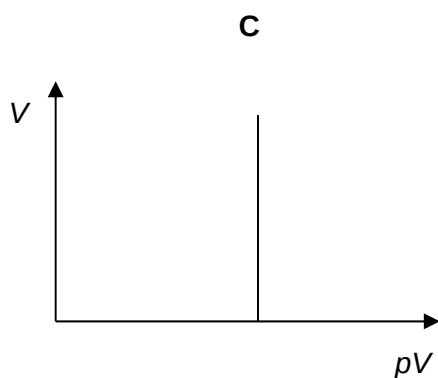
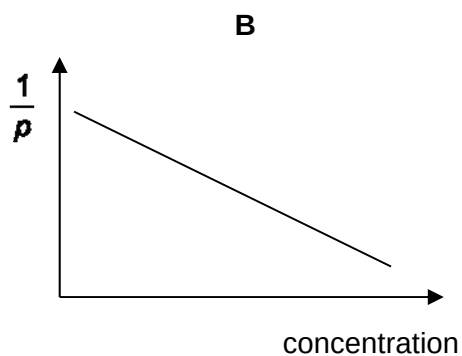
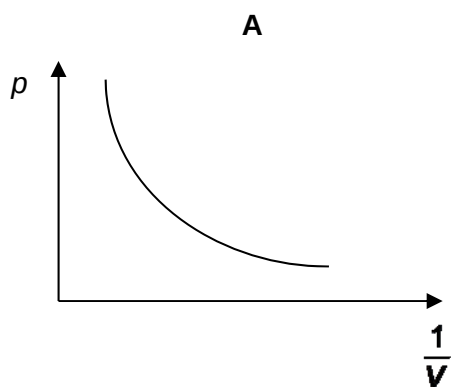
In the structure of solid SiO_2

- each silicon atom is bonded to x oxygen atoms,
- each oxygen atom is bonded to y silicon atoms,
- each bond is a z type bond.

What is the correct combination of x , y and z in this statement?

	x	y	z
A	2	1	ionic
B	2	1	covalent
C	4	2	ionic
D	4	2	covalent

- 7 Which of the following diagrams correctly describes the behavior of a fixed mass of an ideal gas at constant T ?



- 8 In the table below,

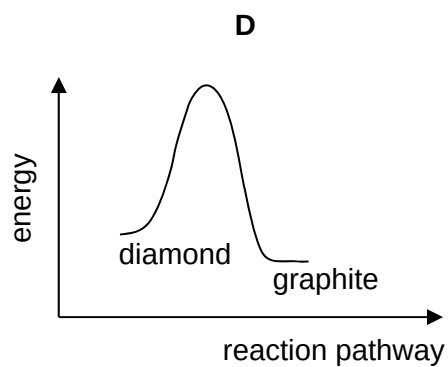
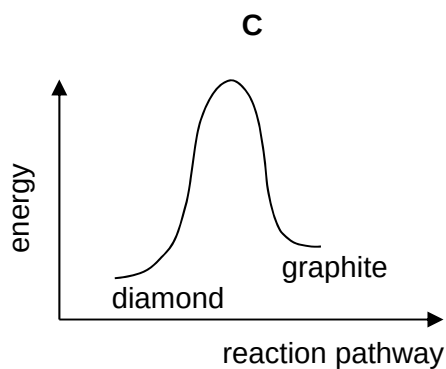
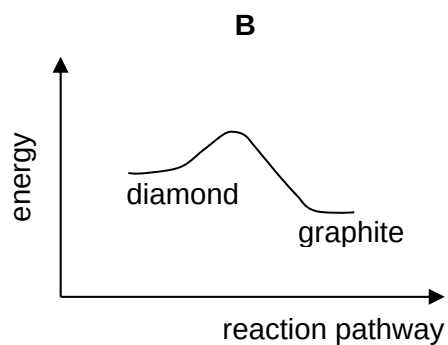
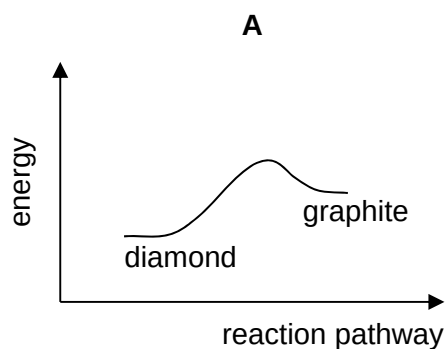
- '+' means that this type of standard enthalpy change can only have positive values,
- '-' means that this type of standard enthalpy change can only have negative values,
- '+ / -' means that either positive or negative values are possible.

Which row is correct?

	atomisation	formation	solution
A	+	+ / -	+ / -
B	+	+	+ / -
C	-	+ / -	-
D	-	-	+

- 9 The conversion of diamond into graphite is an exothermic reaction. Diamond does not readily change into graphite.

Which reaction pathway correctly represents this conversion?



- 10 Which one of the following equations represents a reaction that is feasible at all temperatures?

- | | | |
|----------|--|----------------|
| A | $\text{W(s)} \longrightarrow \text{X(s)} + \text{Y(g)}$ | $\Delta H > 0$ |
| B | $2\text{T(g)} + 3\text{U(g)} \longrightarrow 4\text{V(g)}$ | $\Delta H < 0$ |
| C | $\text{R(g)} \longrightarrow 2\text{Q(g)}$ | $\Delta H < 0$ |
| D | $\text{K(g)} + \text{L(g)} \longrightarrow \text{M(g)}$ | $\Delta H > 0$ |

- 11** The Gibbs free energy change of a system determines whether a reaction is spontaneous, while the equilibrium constant indicates the extent of reaction.

What does the following pair of values for a reaction system indicate?

	<i>values</i>
ΔG_f^θ	– 50.8
K_c	5.80×10^8

- A** No reaction
B Position of equilibrium lies to the left
C Some extent of reaction
D Reaction goes to completion
- 12** The table gives the concentrations and pH values of the aqueous solutions of two compounds **D** and **E**. Either compound could be an acid or a base.

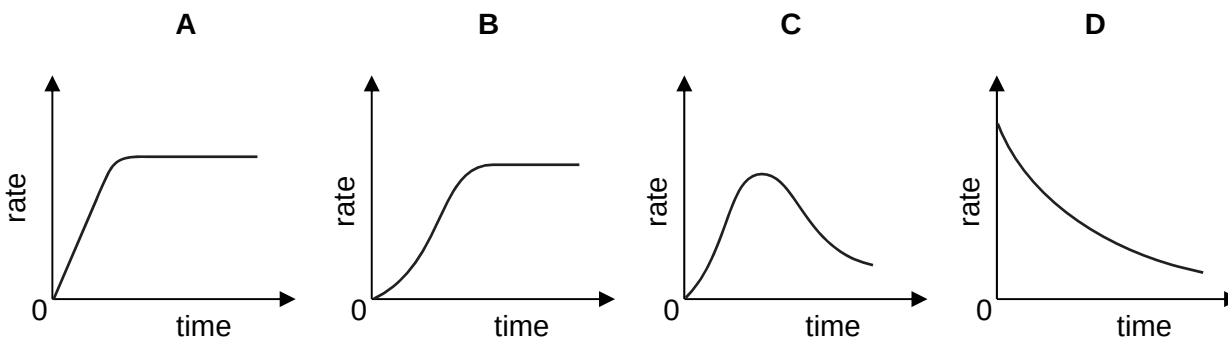
	D	E
concentration / mol dm ⁻³	2.00	2.00
pH	6	9

Student **P** concluded that **D** is a strong acid.

Student **Q** concluded that the extent of dissociation is lower in **E(aq)** than in **D(aq)**.

Which of the students are correct?

- A** both **P** and **Q**
B neither **P** nor **Q**
C **P** only
D **Q** only
- 13** An autocatalytic reaction is a reaction in which one of the products catalyses the reaction.
- Which curve will be obtained if the rate of reaction was plotted against time for an autocatalytic reaction?



14 The information relates to element *W*.

- *W* is in Period 3 of the Periodic Table.
- *W* has a lower electrical conductivity than Mg.
- *W* forms only one chloride which can dissolve in water to give a strongly acidic solution.

What is the likely identity of *W*?

- A** Na **B** Al **C** Si **D** P

15 Consecutive elements **X**, **Y** and **Z** are in the third period of the Periodic Table. Element **Y** has the lowest first ionisation energy and the highest melting point of these three elements.

What could be the identities of **X**, **Y** and **Z**?

- A** phosphorus, sulfur, chlorine
B silicon, phosphorus, sulfur
C magnesium, aluminium, silicon
D aluminium, silicon, phosphorus

16 *Use of the Data Booklet is relevant to this question.*

When 5.00 g of an anhydrous Group II metal nitrate is heated strongly, 3.29 g of gas is produced.

What is the nitrate?

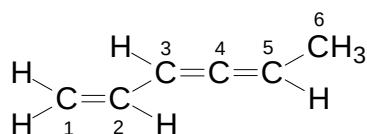
- A** barium nitrate
B calcium nitrate
C magnesium nitrate
D strontium nitrate

17 The great reactivity of fluorine is largely due to the low energy of the F–F bond.

Which statement best accounts for the weak F–F bond?

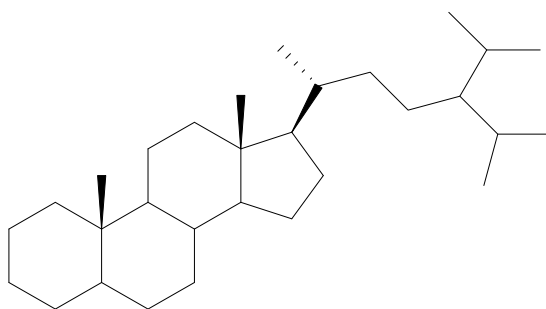
- A** The F–F bond is weak because of repulsion between the non–bonding electrons.
B The F–F bond is weak because of the short length of the bond.
C The F–F bond is weak because of the small nuclear charge of the fluorine atom.
D The F–F bond is weak because of the small size of the fluorine atom.

- 18 Which of the following is true about the compounds of the first row transition metals?
- A $\text{Fe}_2(\text{CO}_3)_3(\text{s})$ can be prepared by reacting $\text{FeCl}_3(\text{aq})$ with $\text{Na}_2\text{CO}_3(\text{aq})$.
- B On addition of $\text{H}_2\text{O}_2(\text{aq})$ to acidified $\text{KMnO}_4(\text{aq})$, the solution remains purple.
- C An orange solution of $\text{K}_2\text{Cr}_2\text{O}_7$ is produced when $\text{NaOH}(\text{aq})$ is added to $\text{K}_2\text{CrO}_4(\text{aq})$.
- D The addition of concentrated HCl to $\text{CuSO}_4(\text{aq})$ produces a yellow solution of H_2CuCl_4 .
- 19 Covalent bonds are formed by orbitals overlap. The shape of unsaturated hydrocarbon molecules can be explained in terms of hybridisation of orbitals.



Which of the following statements does **not** describe the molecule above?

- A The σ bond between C1–C2 is formed by 2sp^2 – 2sp^2 overlap.
- B The σ bond between C2–C3 is stronger than that between C5–C6.
- C The π bond between C4–C5 is formed by 2sp – 2sp^2 overlap.
- D The σ bond between C6–H is formed by 2sp^3 – 1s overlap.
- 20 The molecule 2,4-isopropylcholestane, which has been isolated from a class of sponge, can serve as a biomarker and has determined the first evolutionary appearances of some species.

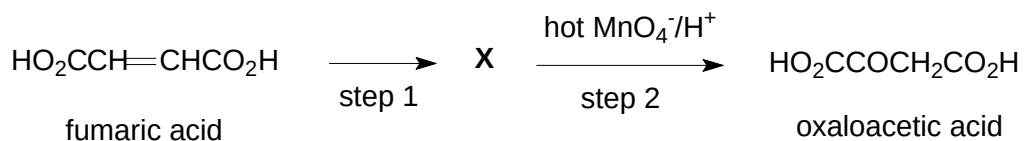


Carbon atoms in a molecule are classified as primary, secondary, tertiary or quarternary, depending on whether they are directly bonded to one, two, three or four carbon atoms.

How many tertiary carbons and how many chiral carbons are there in this molecule?

	tertiary carbons	chiral carbons
A	9	9
B	9	8
C	11	9
D	11	8

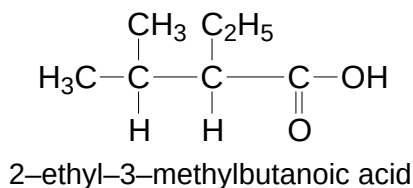
- 21 In the Krebs cycle, fumaric acid is converted to oxaloacetic acid by a two-step process involving an intermediate compound X.



What could be the reagent for step 1 and the intermediate X?

	reagent for step 1	intermediate X
A	cold acidified KMnO_4	$\text{HO}_2\text{CCH}(\text{OH})\text{CH}(\text{OH})\text{CO}_2\text{H}$
B	hot acidified KMnO_4	$\text{OHCCH}(\text{OH})\text{CH}_2\text{CHO}$
C	steam and concentrated H_2SO_4	$\text{HO}_2\text{CCH}(\text{OH})\text{CH}_2\text{CO}_2\text{H}$
D	warm acidified $\text{K}_2\text{Cr}_2\text{O}_7$	$\text{OHCCH}(\text{OH})\text{CH}_2\text{CHO}$

- 22 The characteristic odour of rum is attributed to the compound 2-ethyl-3-methylbutanoic acid.



Which compound, upon heating under reflux with alcoholic sodium cyanide and subsequent acid hydrolysis of the reaction product, will produce 2-ethyl-3-methylbutanoic acid?

- A $\begin{array}{c} \text{CH}_3 \\ | \\ \text{H}_3\text{C}-\text{CH}-\text{CHBr}-\text{CH}_2-\text{CH}_3 \end{array}$
- B $\begin{array}{c} \text{CH}_3 \\ | \\ \text{H}_3\text{C}-\text{CH}-\text{CH}_2-\text{CHBr}-\text{CH}_3 \end{array}$
- C $\begin{array}{c} \text{CH}_3 \quad \text{C}_2\text{H}_5 \\ | \quad | \\ \text{H}_3\text{C}-\text{CH}-\text{CH}-\text{CH}_2\text{Br} \end{array}$
- D $\begin{array}{c} \text{CH}_3 \quad \text{CH}_2\text{Br} \\ | \quad | \\ \text{H}_3\text{C}-\text{CH}-\text{CH}_2-\text{CH}-\text{CH}_3 \end{array}$

- 23 Four isomers of $C_3H_6OCl_2$ are separately subjected to hydrolysis using aqueous sodium hydroxide.

Which isomer produces an organic product with a different molecular formula from the other three isomers?

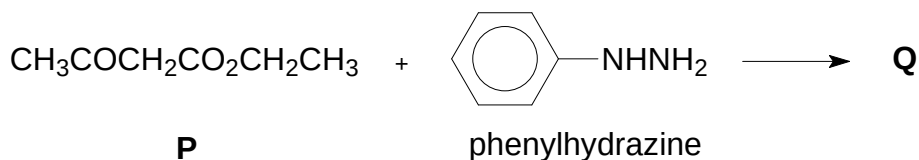
- A $CH_3CCl_2CH_2OH$
- B $CH_2ClCHClCH_2OH$
- C $CH_2ClCH_2CH(OH)Cl$
- D $CHCl_2CH_2CH_2OH$

- 24 Ozone depletion potential (ODP) is a measure of the effectiveness of chlorofluoroalkanes in destroying stratospheric ozone.

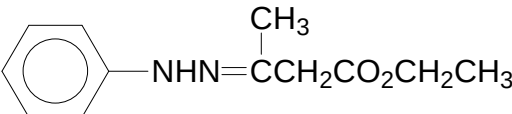
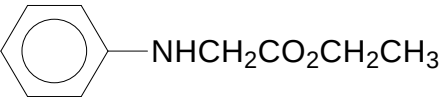
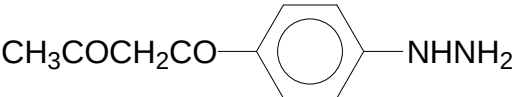
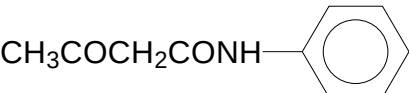
In which sequence are compounds listed in increasing order of their ODPs?

- A $CHClF_2 < CH_3CCl_2F < CCl_2FCClF_2$
- B $CHClF_2 < CCl_2FCClF_2 < CH_3CCl_2F$
- C $CCl_2FCClF_2 < CHClF_2 < CH_3CCl_2F$
- D $CH_3CCl_2F < CCl_2FCClF_2 < CHClF_2$

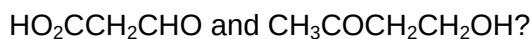
- 25 The first stage in the synthesis of antipyrine, a drug used in reducing fever, is the reaction between compound **P** and phenylhydrazine.



What is the product **Q** of this first stage?

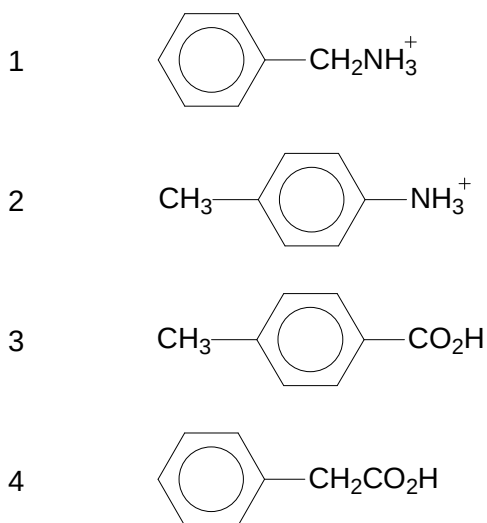
- A 
- B 
- C 
- D 

26 Which reagent could be used to distinguish between



- A acidified potassium dichromate(VI)
- B dilute sulfuric acid
- C Fehling's solution
- D anhydrous PCl_5

27 Consider the following four compounds.



What is the relative order of **decreasing** pK_a ?

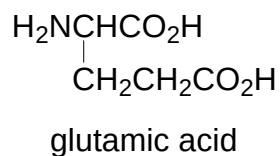
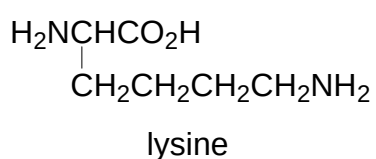
- | | | | | |
|---|---|---|---|---|
| A | 1 | 4 | 3 | 2 |
| B | 4 | 1 | 2 | 3 |
| C | 4 | 1 | 3 | 2 |
| D | 1 | 2 | 4 | 3 |

28 The planet Jupiter is known to have a reducing atmosphere consisting mainly of methane and ammonia, with some ethene and water in the form of ice crystals. Astronomers used the impact of fragments of Comet Shoemaker–Levy with Jupiter to identify more complex molecules present in the planet's atmosphere.

Which molecule is **unlikely** to have been observed?

- A $\text{CH}_3\text{CO}_2\text{H}$
- B $\text{CH}_3\text{CH}_2\text{OH}$
- C $\text{CH}_3\text{CH}_2\text{NH}_2$
- D $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$

- 29 The amino acids lysine and glutamic acid can react with each other to form amide linkages.



What is the maximum number of different compounds, each containing one amide linkage, that can be formed from one molecule of lysine and one molecule of glutamic acid?

- A 2 B 3 C 4 D 5
- 30 A polypeptide undergoes enzymatic hydrolysis at the N-terminal of asp and the C-terminal of gly. The following fragments are obtained.

fragment 1:	asp-gly
fragment 2:	lys-tyr
fragment 3:	asp-cys
fragment 4:	met

Which one of the following is **not** a possible sequence of this peptide?

- A asp-gly-lys-tyr-met-asp-cys
 B lys-tyr-asp-cys-asp-gly-met
 C met-asp-cys-asp-gly-lys-tyr
 D met-asp-gly-lys-tyr-asp-cys

Section B

For each of the question in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct.)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

31 *Use of the Data Booklet is relevant to this question.*

Iron and calcium are metals that form many compounds containing a M^{2+} ion.

Which statement about the electron arrangements in these atoms and ions are correct?

- 1** A Fe^{2+} ion has one less electron shell than a Ca atom.
- 2** A Fe^{2+} ion has the same number of occupied electron shell as a Ca^{2+} ion.
- 3** A valence electron in Fe^{2+} ion experiences the same screening effect as that in Ca^{2+} ion.

32 *Use of the Data Booklet is relevant to this question.*

Adding $KI(aq)$ to a solution containing $Cu^{2+}(aq)$ produces purple–brown iodine solution and a white precipitate of copper(I) iodide, CuI .

Which statements are correct?

- 1** The standard cell potential for the reaction is -0.39 V .
- 2** The production of insoluble CuI reduces the concentration of $Cu^{+}(aq)$ ions from the solution, favouring the forward reaction of the reduction process.
- 3** Addition of sodium thiosulfate to the reaction mixture will cause the white precipitate to dissolve.

- 33** The solubility product, K_{sp} , of calcium phosphate, $\text{Ca}_3(\text{PO}_4)_2$, is **S**.

Which of the following statements about a solution saturated with calcium phosphate are true?

- 1** The concentration of $[\text{Ca}^{2+}]$ at equilibrium is $(\frac{9}{4}\text{S})^{\frac{1}{5}}$.
- 2** The concentration of $[\text{PO}_4^{3-}]$ at equilibrium is $(\frac{2}{108}\text{S})^{\frac{1}{5}}$.
- 3** Addition of NaOH to the equilibrium will increase the solubility product of calcium phosphate.

- 34** $\text{Na}_2\text{S}_2\text{O}_3$ reacts with dilute HCl to give a pale yellow precipitate.

If 1 cm^3 of 0.1 mol dm^{-3} HCl is added to 10 cm^3 of 0.02 mol dm^{-3} $\text{Na}_2\text{S}_2\text{O}_3$, the precipitate forms slowly.

If the experiment is repeated with 1 cm^3 of 0.1 mol dm^{-3} HCl and 10 cm^3 of 0.05 mol dm^{-3} $\text{Na}_2\text{S}_2\text{O}_3$, the precipitate forms more quickly.

Why is this so?

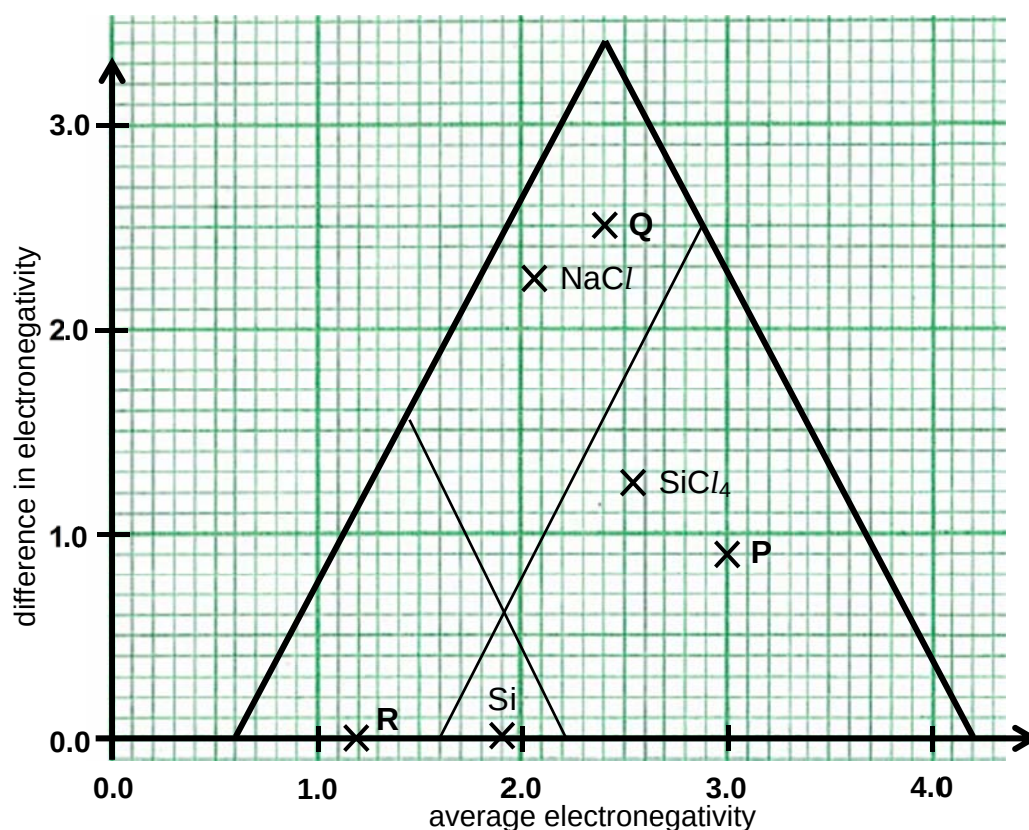
- 1** The reactant particles collide more frequently when 0.05 mol dm^{-3} $\text{Na}_2\text{S}_2\text{O}_3$ is used.
- 2** The activation energy of the reaction is lower when 0.05 mol dm^{-3} $\text{Na}_2\text{S}_2\text{O}_3$ is used.
- 3** The collisions between reactant particles are more violent when 0.05 mol dm^{-3} $\text{Na}_2\text{S}_2\text{O}_3$ is used.

- 35 The van Arkel triangle is used to predict the type of bonding (covalent, ionic or metallic) present in a substance, based on their electronegativity values.

The difference in electronegativity between the element(s) is plotted along the y -axis and their average electronegativity is plotted along the x -axis.

The electronegativity values for some elements are provided in the table below and some of the Period 3 elements and their compounds have been plotted on the van Arkel triangle.

element	electronegativity
Na	0.93
Si	1.90
Cl	3.16



Which of the following substances are non-conductors of electricity?

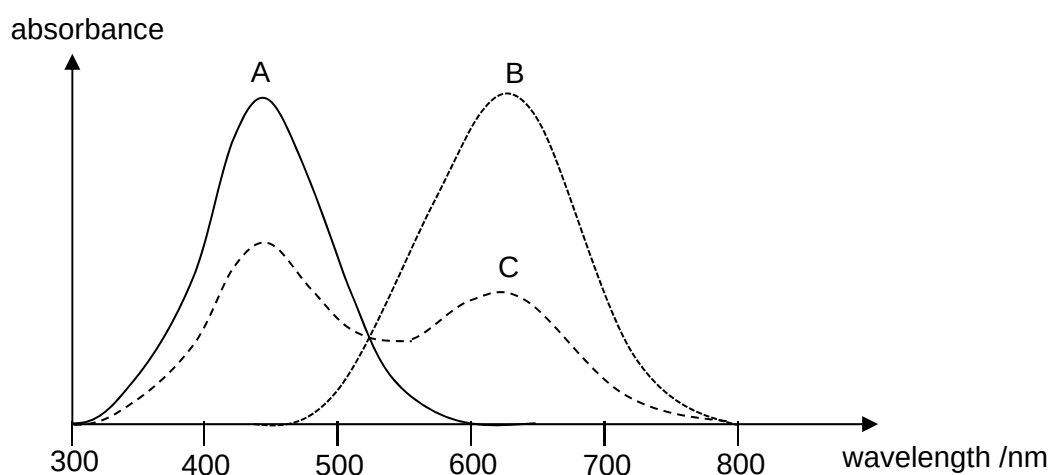
- 1 gas P
- 2 solid Q
- 3 solid R

- 36 When an aqueous solution of a chromium salt is separately mixed with an aqueous solution of ligand *L* and *J*, $[\text{CrL}_6]^{3+}(\text{aq})$ and $[\text{CrJ}_6]^{3+}(\text{aq})$ are formed respectively.

Solutions **A**, **B** and **C** were made up by mixing 0.1 mol dm^{-3} solutions of Cr^{3+} , *L* and *J*. The table below gives the volumes of each used.

solution	volumes of 0.1 mol dm^{-3} solution / cm^3		
	$\text{Cr}^{3+}(\text{aq})$	<i>L</i>	<i>J</i>
A	2	98	0
B	2	0	98
C	2	49	49

The absorbance of a solution at a particular wavelength is proportional to the concentration of the ion responsible for the absorption. The visible absorption spectra of the three solutions **A**, **B** and **C** are shown below.



The table below shows the corresponding colours at the various wavelengths.

wavelength / nm	400	450	500	550	600	650
colour	violet	blue	green	yellow	orange	red

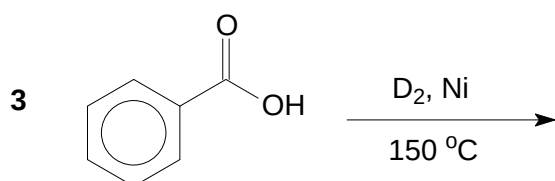
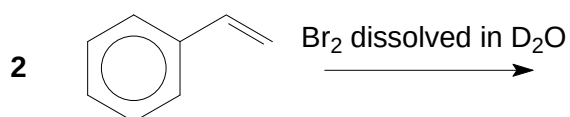
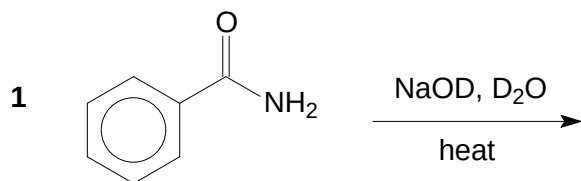
Which of the following statements are correct about the complexes $[\text{CrL}_6]^{3+}(\text{aq})$ and $[\text{CrJ}_6]^{3+}(\text{aq})$?

- 1 $[\text{CrL}_6]^{3+}(\text{aq})$ is blue and $[\text{CrJ}_6]^{3+}(\text{aq})$ is orange-red.
- 2 *L* forms a stronger bond with Cr^{3+} compared to *J*.
- 3 The size of the energy gap between the d-orbitals is bigger in $[\text{CrL}_6]^{3+}$.

37 Which of the following reagents will react with phenol with a colour change being observed?

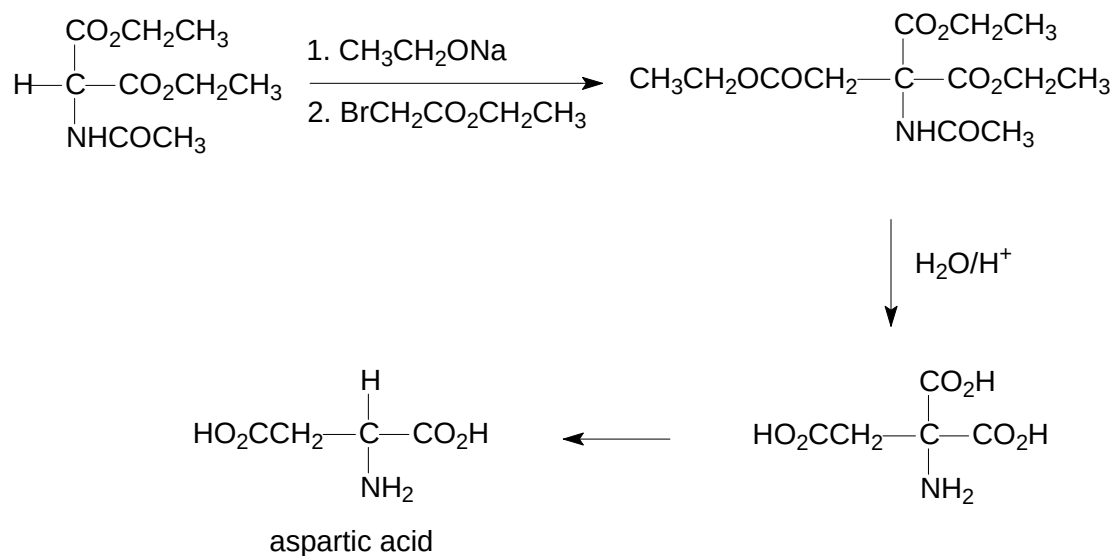
- 1 liquid Br₂
- 2 dilute HNO₃
- 3 aqueous NaOH

38 Which of the following will yield an organic compound containing deuterium, D?
(D = ²H)



39 A general method for the preparation of α -amino acids is the amidomalonate synthesis.

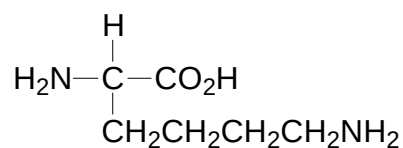
Below is an example of the synthesis of aspartic acid using this method.



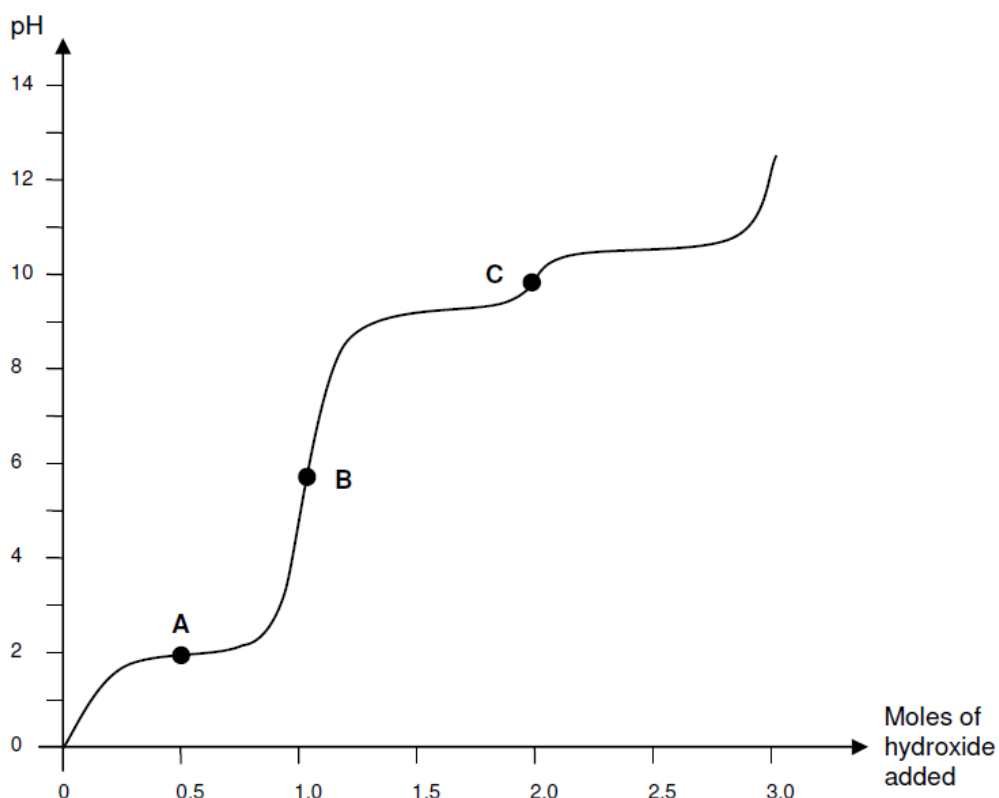
What are the types of reactions shown in this reaction scheme?

- 1 nucleophilic substitution
- 2 acid-base reaction
- 3 hydrolysis

- 40 Lysine is an essential amino acid found in the body. The three pK_a values associated with it are 2.2, 9.0 and 10.5



When 1 mole of protonated lysine was titrated against hydroxide ions, the following pH curve was obtained.



Which of the following statements are true with respect to the curve above?

- 1 The species present at point **C** has no net charge.
- 2 The species present at point **B** will migrate towards the cathode of an electrolytic cell.
- 3 Equal amounts of $\text{H}_3\text{N}^+\text{CH}(\text{CO}_2\text{H})(\text{CH}_2)_4\text{NH}_3^+$ and $\text{H}_3\text{N}^+\text{CH}(\text{CO}_2\text{H})(\text{CH}_2)_4\text{NH}_2$ are present at point **A**

<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	B	21	C
2	A	22	A
3	C	23	B
4	B	24	A
5	D	25	A
6	D	26	C
7	C	27	D
8	A	28	A
9	D	29	D
10	C	30	A
11	D	31	B
12	B	32	B
13	C	33	D
14	C	34	D
15	A	35	B
16	B	36	C
17	D	37	B
18	D	38	C
19	C	39	A
20	B	40	B

A	9
B	11
C	10
D	10